

Migrating to Azure Local



Agenda

- **Objective:** Explore end-to-end VM migration to Azure Local by using Azure Migrate
- **Scope:** Migration lifecycle (discovery, assessment, replication, cutover), Hyper-V/VMware support, networking, Azure Arc, security, monitoring, automation, and troubleshooting.



Introduction to Migrating to Azure Local

- **Migration to Azure Local:** Overview of strategies and tools for moving VM workloads.
- **Supported Platforms:** Hyper-V, VMware, and other on-premises environments.
- **First-Party & Third-Party Options:** Microsoft and partner solutions.
- **Key Considerations:** Planning, prerequisites, and supported configurations.
- **End-to-End Process:** Discovery, replication, migration, and validation.

Migration Options Overview

- **First-Party Migration:**
 - Azure Migrate: Central hub for discovery, assessment, and migration.
 - No additional cost for built-in tools.
- **Third-Party Migration:**
 - Carbonite, Commvault, Veeam: Partner solutions for specialized needs.
- **Selection Criteria:**
 - Current infrastructure compatibility.
 - Migration goals and complexity.
 - Support for automation and reporting.

Azure Migrate

- **Azure Migrate:**
 - **Discovery:** Identifies on-premises VMs and dependencies.
 - **Assessment:** Evaluates readiness and sizing for Azure Local.
 - **Migration:** Orchestrates replication and cutover.
- **Supported VM Types:**
 - Hyper-V VMs (Preview).
 - VMware VMs.
- **Arc-Enabled VMs:**
 - All migrated VMs are **Azure Arc-enabled** by default.

Azure Migrate Architecture and Components

- **Azure Layer:**
 - **Azure Migrate Project:** Orchestrates discovery and replication.
 - **Storage Accounts & Key Vault:** Store metadata and replication state.
 - **Azure Resource Manager (ARM):** Manages resources across Azure and Azure Local.
- **On-Premises Layer:**
 - **Source Servers:** VMware/Hyper-V hosts.
 - **Source Appliance:** Discovers and prepares VMs.
 - **Target Appliance:** Receives migrated VMs on Azure Local.
 - **Arc Resource Bridge & Failover Cluster Manager:** Orchestration and provisioning.

Azure Migrate Migration Phases

- **1. Prepare:**
 - Complete prerequisites.
 - Deploy and register Azure Local.
 - Create Azure Migrate project and storage account.
- **2. Discover:**
 - Configure source appliance.
 - Discover on-premises VMs.
- **3. Replicate:**
 - Configure target appliance.
 - Select and replicate VMs.
- **4. Migrate & Verify:**
 - Migrate VMs to Azure Local.
 - Validate boot and data integrity.
 - Decommission source VMs.

Supported Configurations and Requirements

- **Source Environments:**
 - Hyper-V on Windows Server 2012 R2–2025.
 - VMware vCenter/ESXi 6.5–8.0.
- **Appliance OS:**
 - Windows Server 2022 (source and target).
- **Target Environment:**
 - Azure Local 2311.2 or later.
- **Guest OS Support:**
 - <https://learn.microsoft.com/en-us/windows-server/virtualization/hyper-v/supported-windows-guest-operating-systems-for-hyper-v-on-windows>.
- **Geographies:**
 - Asia-Pacific, Europe, United States (metadata storage).

Roles, Provider Registration, and Project Considerations

- **Roles:**
 - Entra tenant: Application Developer.
 - Azure subscription: Contributor, User Access Administrator.
- **Resource Provider Registration:**
 - Microsoft.DataReplication must be registered.
- **Project and Appliance Pairing:**
 - One source and one target appliance per Azure Migrate project.
 - Separate projects for Hyper-V and VMware sources.

Source and Target Prerequisites

- **Source (Hyper-V/VMware):**
 - Sufficient resources for appliance VM (16 GB RAM, 80 GB disk, 8 vCPUs).
 - VMs must be highly available (for clustered hosts).
 - Disks must be attached to CSV (Hyper-V).
 - BitLocker and encrypted disks must be disabled.
 - No shared disks or Azure Connected Machine Agent.
- **Target (Azure Local):**
 - Azure Arc resource bridge deployed.
 - Logical network and custom storage path configured.

Azure Migrate Project Setup

- **Create Project in Azure Portal:**
 - Select subscription and resource group.
 - Name the project and choose geography.
- **Discovery and Replication:**
 - Start discovery from the Azure Migrate dashboard.
 - Configure source and target appliances.
 - Review discovered VMs and initiate replication.

Migration Process

- **Discovery:**
 - Install and configure source appliance.
 - Power on VMs and ensure integration services/tools are installed.
- **Replication:**
 - Generate project keys for source and target appliances.
 - Download and install appliances (.VHD, .OVA, or .zip).
 - Register appliances and validate connectivity.
- **Migration:**
 - Select VMs, configure compute and storage settings.
 - Start migration and monitor progress.
 - Complete migration and clean up resources.

Network Prerequisites: Ports

- **Firewall Ports (Hyper-V source / Azure Local target)**
 - 3389 (RDP): Appliance remote administration.
 - 44368 (Appliance Manager): Access via `https://<appliance-ip-or-name>:44368`.
 - 5985 / 5986 (WinRM): Appliance-to-host communication.
 - 445 (SMB): Source ↔ Target appliance data movement.
- **Firewall Ports (VMware source specifics)**
 - 443 (Azure services & vCenter): Orchestrates **replication/migration** and **vCenter** connectivity.
 - 902 (ESXi heartbeat/snapshots): Data from ESXi snapshots; heartbeat to vCenter.

Network Prerequisites: URLs & General Guidance

- **Required URLs**
 - Azure Migrate appliance URL access.
 - *.siterecovery.azure.com for orchestration.
- **General Guidance**
 - Confirm **bidirectional** flows where noted.
 - Avoid overlapping **snapshot schedules** during replication.

Source Hyper-V Requirements & Constraints

- Supported Hyper-V Hosts: Windows Server 2012 R2 → 2025.
- Discovery Scope
 - Standalone host VMs are discoverable/migratable.
 - Clustered hosts: Only HA VMs are eligible; non-HA VMs on clusters must be made HA first.
- Appliance Capacity (Source)
 - Create a Windows Server 2022 VM: 16 GB RAM, 80 GB disk, 8 vCPU.
- Disk Preconditions
 - CSV-attached disks required for migration.
 - DAS/SMB shares on source not supported.
- Guest Prep
 - Bring disks online; persist drive letters (SAN policy).
 - Uninstall Azure Connected Machine Agent if present.
 - Disable BitLocker; decrypt encrypted volumes; no shared disks.

Target Azure Local Requirements & Setup

- **Azure Local Version: 2311.2 or later** for target migrations.
- **Azure Arc Resource Bridge (ARB)**
 - Must exist; created during deployment; verify via portal.
- **Logical Network**
 - Create and select a logical network for migrated VMs.
- **Custom Storage Path**
 - Configure storage path on ARB; ensure capacity for replicated disks.
- **Connectivity**
 - Source ↔ Target must reach each other (same network or VPN).

Azure Migrate Project Rules & Geographies

- **1:1 Appliance Pairing**
 - One source appliance + one target appliance per project.
 - Separate projects for Hyper-V vs VMware.
- **Tenant & Subscription Alignment**
 - Project must be in **same tenant** as **Azure Local**.
 - Register **Microsoft.DataReplication** RP on all involved subscriptions.
- **Project Geography**
 - **Metadata** only stored in selected geography/region (e.g., **Central US**, **West US2**; **North/West Europe**; **SE/East Asia**).
 - VM data remains **on-prem**; **no customer data** leaves allocated region.

Hyper-V: Source Appliance Creation & Discovery

- **Generate Project Key**
 - In Azure Migrate → Inventory → Start discovery → Using appliance → For Azure Local.
- **Install Source Appliance**
 - Template (.VHD) method, or Script (.zip) via AzureMigrateInstaller.ps1.
 - Appliance host: Windows Server 2022, 16 GB RAM, 80 GB disk, 8 vCPU.
- **Configure & Register**
 - Use Appliance Manager; optional preconfigured Entra ID app.
- **Discover VMs**
 - Ensure VMs are powered on with Hyper-V integration services installed; wait for green checkmark.

Hyper-V: Target Appliance Creation & Registration

- **Generate Target Key**
 - In Azure Migrate → Replicate → Deploy and configure target appliance → Generate key.
- **Install Target Appliance**
 - Download .VHD or .zip, validate SHA-256, deploy on Azure Local.
 - VM sizing: Windows Server 2022, 16 GB RAM, 80+ GB disk, 8 vCPU.
- **Register & Validate**
 - Use Azure Migrate Target Appliance Configuration Manager; sign-in, MFA; status Registered/Validated.
- **Provide System Info**
 - FQDN, domain, credentials for the Azure Local system; Configure until Successful..

Hyper-V: Start Replication Wizard Part 1

- **Basics Tab**
 - Select subscription, resource group, Target system (Azure Local).
 - Prereqs show green checks (e.g., Arc Resource Bridge).
- **Target Appliance Tab**
 - Verify connected status (green check).
- **Virtual Machines Tab**
 - Discovered VM list; select up to 10 per batch.
- **Target Settings Tab**
 - Storage account (metadata only): Standard, Public network access enabled.
 - Logical network: pre-created static/dynamic logical network; Reload if missing.
 - Storage path: pre-created on ARB; ensure free capacity; Reload if missing.

Hyper-V: Start Replication Wizard Part 2

- **Compute & Disks Tabs**
 - Rename target VMs, assign OS disk, set vCPU/RAM; choose **disks to replicate** (OS disk mandatory).
- **Review + Start**
 - Verify all values; **Replicate**; stay on page until artifacts finalize.

Hyper-V: Migrate & Verify; Complete Migration

- **Migrate VMs**
 - In **Replications**, choose **Migrate** (bulk or single VM).
 - Optionally **shutdown** source VMs to avoid data loss.
- **Verify Post-Cutover**
 - In **Azure Local** → **Virtual machines**, confirm **Running** state; verify **disks & NICs**; test via **VMConnect/apps**.
- **Complete Migration (Cleanup)**
 - In portal: **Complete migration**; removes **protected items**, **seed artifacts** (**seed.iso**, **seed disks**).
 - Frees storage; allows **re-protect** later if needed.

VMware: Source Appliance Creation & Discovery

- **Generate Project Key**
 - Azure Migrate → Inventory → Start discovery → Using appliance → For Azure Local → VMware.
- **Install Source Appliance**
 - .OVA deployment or script (.zip) on Windows Server 2022 VM (16 GB / 80 GB / 8 vCPU).
 - Install VMware VDDK (8.0.0/8.0.1/8.0.2; avoid 7.0.x).
- **Register & Configure**
 - Optional preconfigured Entra ID app; add vCenter credentials; add discovery sources.
 - Disable agentless application discovery (not supported for Azure Local migrations).
- **Discover VMs**
 - Ensure VMware Tools installed; VMs powered on; wait for green checkmark.

VMware: Replicate, Migrate & Verify; Complete Migration

- **Start Replication**

- Use **Replicate** wizard: **Basics** / **Target appliance** / **Virtual machines** / **Target settings** / **Compute** / **Disks** / **Review + Start**.
- **Storage account** is for **metadata only**; ensure **Public network access**; **Standard tier**.
- Select **logical network** and **ARB storage path**; ensure capacity; **Reload** if not listed.

- **Migrate & Verify**

- Trigger **Migrate** from **Replications**; optionally **shutdown** source VM.
- Validate **Running state**, **disk & NIC config**, and application behavior in **Azure Local**.

- **Complete Migration**

- Use **Complete migration** to remove protected items and leftover **seed artifacts**; free capacity; enable re-protection.

Maintaining Static IPs During Migration

- **Two Environment Types**
 - SDN-enabled Azure Local → static IPs preserved automatically.
 - Non-SDN Azure Local → static IP migration requires scripts / packages.
- **Why IP Preservation Matters**
 - Ensures application continuity.
 - Prevents breaking DNS, application bindings, firewall rules, legacy integrations.

Static IP Preservation in SDN-Enabled Azure Local

- **Automatic Preservation**
 - Static IPs retained when assigned to **static logical networks**.
- **Key Considerations**
 - Multi-NIC VMs: SDN assigns **default gateway** to all NICs → remove gateway on secondary NICs.
- **SDN Documentation References**
 - SDN Integration (Azure Arc).
 - Static Logical Networks configuration guidance.

Static IP Preservation in Non-SDN Azure Local (Windows)

- **Windows Support**
 - Requires Static IP Migration Package (.zip).
- **Included Scripts**
 - Prepare-MigratedVM.ps1
 - Initialize-StaticIPMigration.ps1
 - Set-StaticIPConfiguration.ps1
 - Utilities.psm1
- **Execution Flow**
 - Capture IP configuration pre-migration.
 - Store config in **InterfaceConfigurations.json**.
 - Apply configuration on first boot in Azure Local.
- **Pre-Migration Requirements**
 - VM must remain **powered on**.
 - Admin account required for scheduled tasks.

Static IP Preservation in Non-SDN Azure Local (Linux)

- **Linux Support**
 - Requires Static IP Migration Package (.zip).
- **Included Scripts**
 - `prepare_migrated_vm.sh`
 - `azure_local_ip_helper.py`
 - `static_ip_manager.sh`
- **Execution Flow**
 - Capture IP configuration pre-migration via the `./prepare_migrated_vm.sh -o StaticIPMigration` command
 - Store config in a localized guest JSON file (`network_state.json`)
 - Apply configuration on first boot via a persistent systemd startup service.
- **Pre-Migration Requirements**
 - VM must remain **powered on**.
 - Root or sudo privileges required to inject startup binaries
 - python3 package must be installed for configuration parsing

Preparing VMs for Static IP Migration

- **Source VM Requirements**
 - VMware Tools or Hyper-V Integration Services installed.
 - VM must stay **powered on** through replication.
 - Admin credentials required to install scripts.
- **Logical Network Requirements**
 - Static IPs must be **free** in target network pools.
 - Target logical network must match **subnet layout**.
- **Common Failure**
 - If IP is already assigned → Migration fails with “address already in use.”

Monitoring Migrations with Azure Migrate

- **Diagnostic Logs**
 - Enable logging on the **vault** associated with the Azure Migrate project
 - Supported categories: **ProtectedItems**, **Jobs**, **Errors**, more.
- **Destinations**
 - **Log Analytics Workspace** (query & dashboards).
 - **Azure Storage Account** (archival).
- **Why Enable Diagnostics**
 - Troubleshooting replication cycles.
 - Audit compliance & historical analysis.
 - Track failures across appliances.

Log Analytics for Migration Visibility

- Example Kusto Query

```
ASRv2ProtectedItems  
| where TimeGenerated > ago(30d)
```

- Common Queries

- Identify failing replications.
- Track planned failovers (migrations).
- Correlate appliance warnings and errors.

- Recommended Dashboards

- Migration readiness overview.
- Replication health over time.

Enabling Guest Management on Migrated VMs

- **Guest Agent Requirements**
 - Azure Local VM must run **Windows Server 2016+** or supported Linux distribution.
 - **Public network connectivity** required.
- **Generation-Specific Requirements**
 - **Gen1 VMs must be powered off** before enabling guest agent.
 - Gen2 can be powered on or off.
- **Activation Flow**
 1. Attach **guest agent ISO**.
 2. Run **install script** (Windows or Linux).
 3. Enable guest management from **Azure CLI or portal**.

PowerShell Automation for Migration

- **Supported Automation Areas**
 - Retrieve discovered servers.
 - Initialize replication infrastructure.
 - Create replication jobs.
 - Update and modify protected items.
 - Trigger migration (planned failover).
 - Complete migration and remove protection.
- **Key Cmdlets**

Get-AzMigrateDiscoveredServer

Initialize-AzMigrateLocalReplicationInfrastructure

New-AzMigrateLocalServerReplication

Set-AzMigrateLocalServerReplication

Start-AzMigrateLocalServerMigration

Remove-AzMigrateLocalServerReplication

Troubleshooting Migration Failures

- **Frequent Root Causes**

- Appliance registration issues.
- Insufficient storage or full CSV.
- VMs not highly available (Hyper-V clusters).
- Disk discovery failures ("value cannot be null").
- IP conflicts during static IP migration.
- Snapshot failures (VMware / Hyper-V).

- **Troubleshooting Tools**

- Diagnostic logs.
- Appliance logs under ProgramData.
- Azure portal Job IDs & correlation IDs.
- Manual commands (e.g., Get-Service, snapshot operations).

Troubleshooting: Appliance Registration Failures

- **Common Causes**

- Incorrect **FQDN resolution** from appliance.
- Missing **WinRM configuration** on Azure Local nodes.
- Not using a **preconfigured Entra ID app** where required.

- **Fixes**

- Add target system to **hosts file** manually: IP FQDN
- Run **Enable-PSRemoting -Force** on each target node.
- Re-register using Appliance Manager.

Troubleshooting: Target System Info Cannot Be Modified

- **Cause**

- Target system details stored in **CredStore** cannot be edited once set.

- **Resolution (only before replication begins)**

- Delete:

C:\ProgramData\Microsoft Azure\CredStore\TargetClusterCredentials.json

- Re-open appliance manager and re-enter values.

- **Important Warning**

- Not allowed after replication begins — appliance must be removed & recreated.

Troubleshooting: Replication Fails (Capacity, HA, CSV)

- **Root Causes**

- CSV volume full or insufficient free space.
- VM not **highly available** on Hyper-V cluster.
- OS disk not attached to **Cluster Shared Volume**.

- **Fixes**

- Expand CSV volume as needed.
- Convert VM to **high availability**.
- Verify disk attachment inside Failover Cluster Manager.

Troubleshooting: Snapshot & Checkpoint Failures

- **Symptoms**

- Snapshot creation fails during replication.
- Migration job displays **"Failed to delete snapshot"**.

- **Root Causes**

- Hyper-V host issues retrieving VHD metadata.
- VMware snapshot conflicts or locked VMDK files.

- **Resolution Steps**

- Validate `Get-VM`, `Get-VHD`, and manual checkpoint operations.
- Resolve storage or host errors before retrying replication.

Troubleshooting: “Value Cannot Be Null” Disk Errors

- **Error Message**
 - Value cannot be null. Parameter name: FetchingHyperVDiskPropertiesFailed
- **Cause**
 - Hyper-V host fails to return correct disk metadata (MaxInternalSize, ParentPath).
- **Fix**
 - Run WMI query via:
Get-WmiObject -Class Msvm_ImageManagementService
 - Validate output XML and repair disk chain if needed.

Troubleshooting: VMs Fail to Shut Down During Migration

- Scenario
 - Source VM fails to power off during planned failover.
- Root Cause
 - WMI shutdown component failure.
- Fix

```
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Query "Select * From  
Msvm_ComputerSystem Where Name='VMID' "
```

```
$shutdown = Get-WmiObject -Namespace root\virtualization\v2 -Query  
"Associators of {$vm} Where AssocClass=Msvm_SystemDevice  
ResultClass=Msvm_ShutdownComponent"
```

```
$shutdown.InitiateShutdown("TRUE", "Need to shutdown")
```


Troubleshooting: Windows Server BSOD

- **Scenario**
 - Windows Server 2022+ BSOD after migrating from VMware to Azure Local
- **Root cause**
 - Incorrect VMware storage/NIC drivers after Hyper-V conversion
 - Windows Server 2022+ stricter enforcement of driver validation.
- **Fix**
 - Source VM: Replace **LSI Logic SAS** (SCSI) and **E1000/E1000E** (NIC) with PVSCSI and VMXNET3. Reinstall VMware tools.
 - Azure Local VM: Create as Gen 2 VM, enable Secure Boot, and use Hyper-V synthetic NIC.
 - Post-Migration cleanup: Remove VMware Tools after migration

```
Get-WmiObject Win32_Product |  
Where-Object { $_.Name -like "*VMware*" } | ForEach-Object { $_.Uninstall() }
```

Troubleshooting: Cleaning Up Failed Migrations

- **If failure happens *before* VM creation**
 - No manual cleanup needed.
 - Retry migration.
- **If failure happens *after* VM creation**
 - Delete VM in Azure Local.
 - Validate deletion in Hyper-V Manager.
 - Do **not** delete:
 - Migrated disk(s)
 - Seed disk(s)
 - Network interfaces
 - Azure Migrate reuses these on retry.

Post-Migration: Guest Agent Enablement Issues

- **Gen1 VMs**
 - Must be **powered off** before attaching guest-agent ISO.
- **If ISO attach fails**
 - Symptoms: VM unmanageable in portal; DVD attach errors.
- **Fix**
 - Stop VM via Azure CLI:
az stack-hci-vm stop --name <vmName> --resource-group <rgName>
 - Re-run guest agent enable command.

Final Migration Validation & Cleanup Checklist

- **Confirm VM Boot Success**
 - Validate app startup, OS services, event logs.
- **Verify Network Functionality**
 - Check static IP assignment (if applicable).
 - Test cross-subnet communication.
- **Storage Validation**
 - Confirm correct disks attached.
 - Validate VHD/VMDK size alignment.
- **Cleanup**
 - Complete migration in portal.
 - Remove source VMs from Hyper-V / vCenter.
 - Decommission source & target appliances if no longer needed.

Key Migration Unsupported Scenarios & Constraints

- **Not Supported**
 - VMs with **encrypted disks** (BitLocker or others).
 - **Shared disks** or clustered shared volumes inside guest.
- **Additional Constraints**
 - Only **one source + one target appliance** per project.
 - Can't migrate VMs using **DHCP reservations** expecting MAC preservation.

Post-Migration: DNS & Network Re-Registration

- **DNS Considerations**
 - Ensure updated **A-records** reflect migrated IP.
 - Remove stale DNS entries from source environment.
- **Network Verification**
 - Validate **NIC name changes** after migration.
 - Confirm **default gateway** assignments (especially multi-NIC).
 - Re-apply **custom firewall rules**.
- **Application Dependencies**
 - Test API endpoints, service bindings, SMB shares, LDAP paths.

Post-Migration: Storage & Disk Integrity Validation

- **Disk Checks**
 - Confirm OS & data disks attached as expected.
 - Compare **VHD/VMDK sizes** pre- vs post-migration.
 - Validate disk **online status** & drive letters (Windows).
- **SAN / fstab Considerations**
 - Windows: ensure **SAN policy** = **OnlineAll**.
 - Linux: confirm **persistent mount identifiers**.
- **Common Symptoms**
 - Offline disks after cutover.
 - Incorrect disk mapping due to non-CSV sources.

Migration Cleanup

- **Clean Up Target Environment**
 - Remove **appliance VMs** (source + target).
 - Remove stored **logs** and **seed disks**.
 - Validate no stale NICs remain in logical networks.
- **Clean Up Azure Resources**
 - Delete Azure Migrate project (optional).
 - Delete replication vault metadata.
- **Hyper-V / VMware Source**
 - Decommission migrated VMs.
 - Validate cluster health after removal.

Best Practices for Large-Scale VM Migrations

- **Batch Migration Strategy**
 - Replicate in **10-VM batches** as supported.
 - Stagger cutover windows by workload tier.
- **Operational Readiness**
 - Ensure **monitoring & diagnostics** enabled.
 - Perform baseline checks on appliances weekly.
- **Risk Mitigation**
 - Pre-cutover **snapshot / checkpoint** in source (VMware only).
 - Document **fallback** procedure.

Automation for Enterprise-Scale Cutovers

- **Azure PowerShell Automation Includes**
 - Bulk `Start-AzMigrateLocalServerMigration` calls.
 - Automated validation workflows.
 - Scheduled migration windows.
- **Pipeline Integration**
 - CI/CD orchestration for migration waves.
 - Integration with **Azure DevOps** or **GitHub Actions**.
- **Benefits**
 - Predictability, repeatability, reduced human error.

Common Operational Questions

- Can Windows Server 2008 R2 be migrated?
 - Yes, with SP1 and patch KB3138612 installed.
- Are Secure Boot VMs supported?
 - Yes.
- Can VMware and Hyper-V be migrated in same project?
 - No.

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